

Traditional NLP Baselines vs Gemini

Task 1: Sentiment Analysis (SST-2)

Dataset: SST-2 ([stanfordnlp/sst2](#), Hugging Face)

Data construction (from train split):

The training data was prepared by first selecting the initial 10,000 positive and 10,000 negative sentences from the training set, resulting in a total dataset of 20,000 sentences.

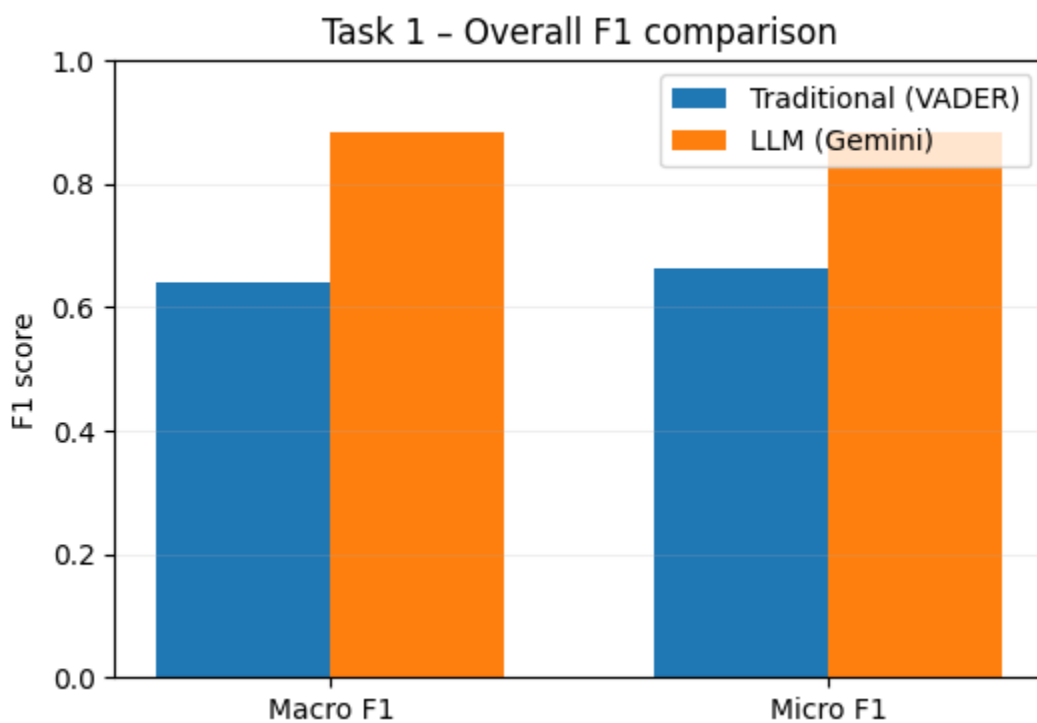
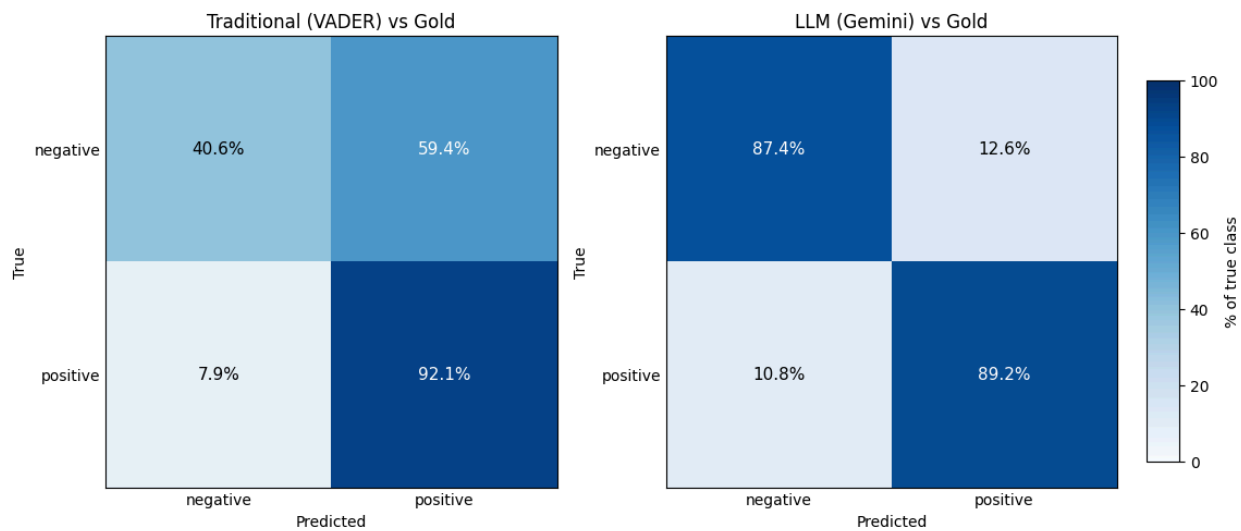
Eg. director peter kosminsky gives these women a forum to demonstrate their acting ` chops ' and they take full advantage . (positive)

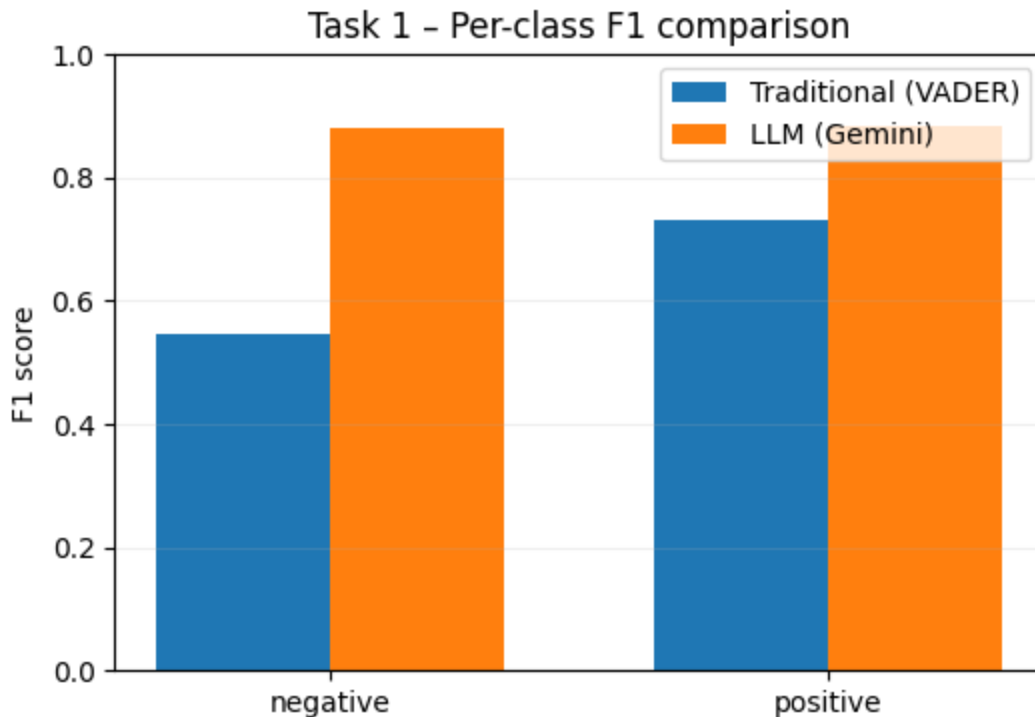
Traditional baseline method:

The sentiment analysis approach uses NLTK VADER's [SentimentIntensityAnalyzer](#) to score each sentence in the text. The VADER compound sentiment score is then computed for the entire text. A prediction of "positive" is assigned if the compound score is greater than or equal to zero, and "negative" is assigned otherwise.

The VADER compound score is a rule-based metric that calculates a text's overall sentiment by summing individual word ratings from a predefined lexicon, adjusting for contextual rules like punctuation and capitalization, and normalizing the total into a single value between -1.0 (most negative) and +1.0 (most positive).

Comparison:





Task 2: Named Entity Recognition (NER) (CoNLL-2003)

Dataset: CoNLL-2003 train data (downloaded from <https://data.deepai.org/conll2003.zip>)

Data construction (from train file):

The process involves parsing the train file in CoNLL-2003 format (where each line contains: word, pos, chunk, ner), splitting it into sentences using blank lines, randomly but deterministically shuffling sentence indices with a fixed seed, selecting 1,000 sentences, and for each selected sentence, building the text by joining tokens with spaces and formatting the gold entities (converted from IOB2 tags PER, ORG, LOC, MISC) as "Entity (TYPE); Entity2 (TYPE)" or "NONE" if no entities exist.

LOC: 491 (30.0%)

ORG: 471 (28.7%)

PER: 446 (27.2%)

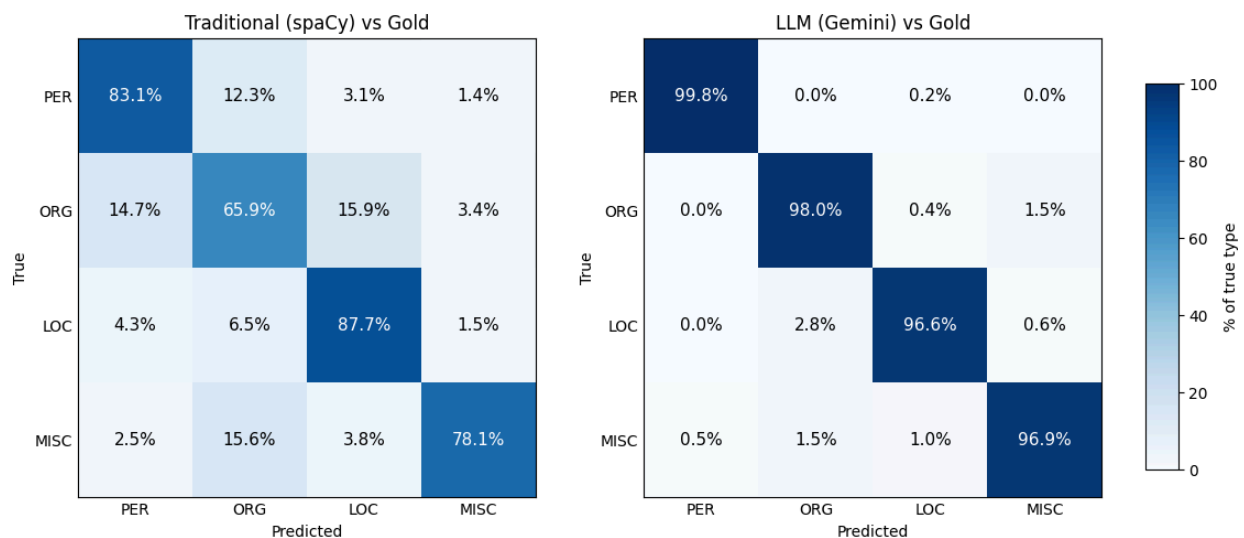
MISC: 231 (14.1%)

Eg. Medicaid would be at risk of being taken away... — Medicaid (MISC)

Traditional baseline method:

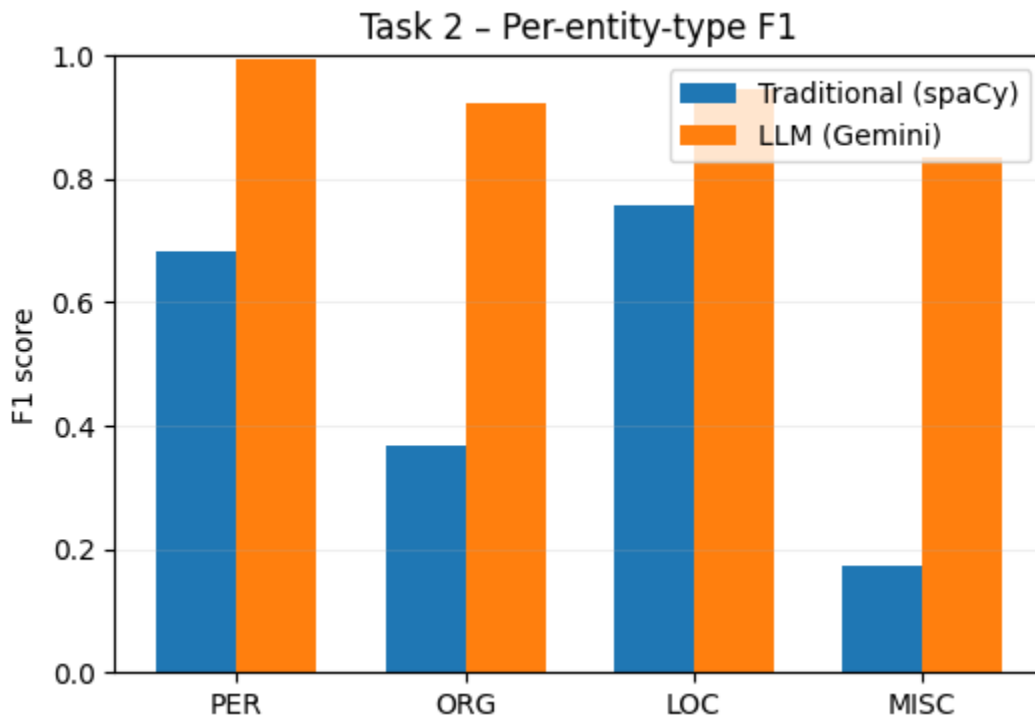
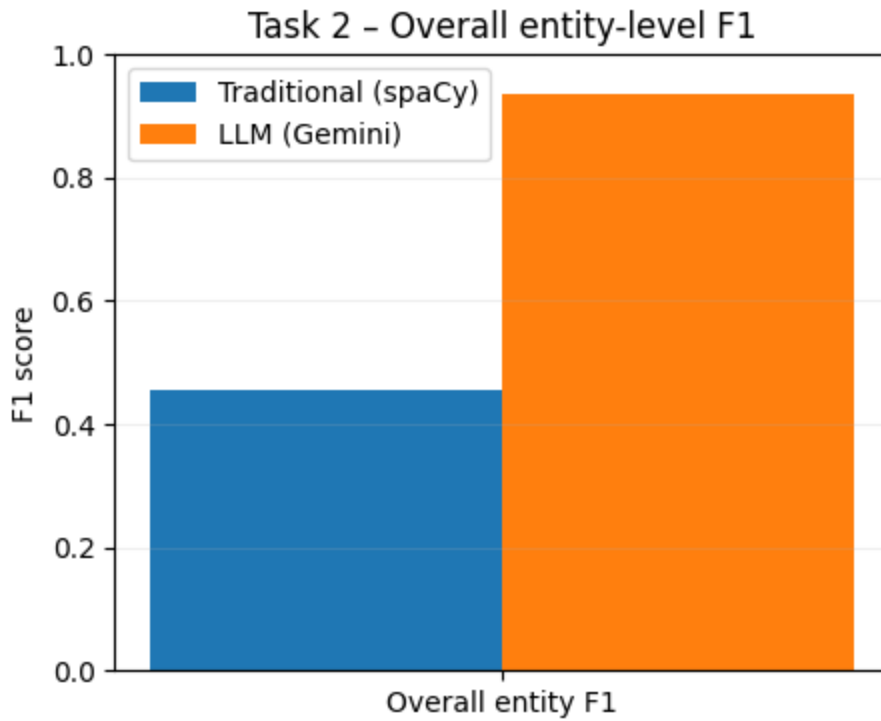
The standard approach for Named Entity Recognition (NER) involves using a library such as **spaCy** with a small English model (e.g., `en_core_web_sm`) to process text. For each input text, the model is run, and the resulting `doc.ents` are extracted to identify predicted entities. These spaCy-detected entity labels must then be mapped to a standardized set of types, like the CoNLL scheme, where, for instance, **PERSON** becomes **PER**, **ORG** remains **ORG**, and location-related labels like **GPE**, **LOC**, and **FAC** are consolidated into **LOC**, with all other types categorized as **MISC**. The final output must present the predicted entities in a specified string format, such as "**Entity (TYPE); ...**" or simply "**NONE**" if no entities are successfully identified.

Comparison:



Traditional (spaCy): 0.4551

LLM (Gemini): 0.9373



Task 3: POS Tagging (CoNLL-2003)

Dataset: The **same 1,000 sentences** sampled for the NER task.

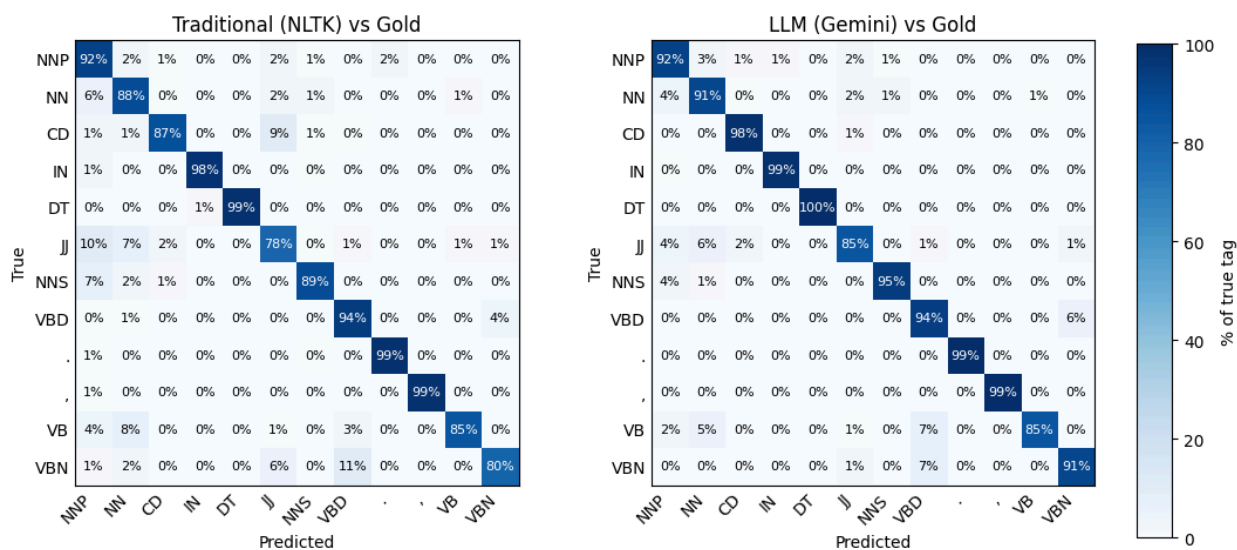
Data construction:

For each selected sentence, the text is built by joining tokens with spaces. The `pos_ground_truth` is built as a single sequence formatted as "w1/T1 w2/T2 ...". The POS tags are sourced directly from the CoNLL POS column.

Traditional baseline method:

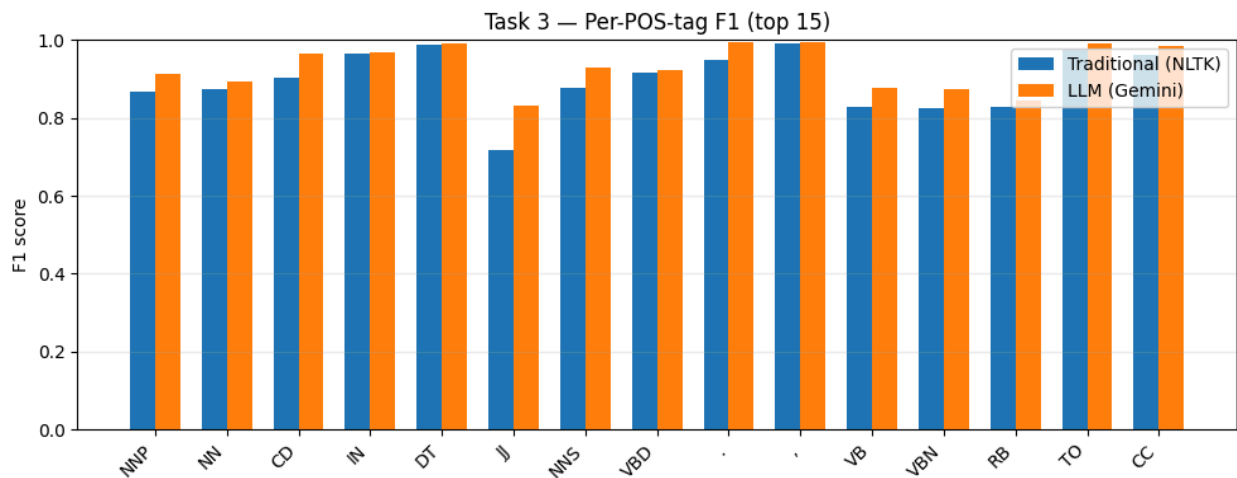
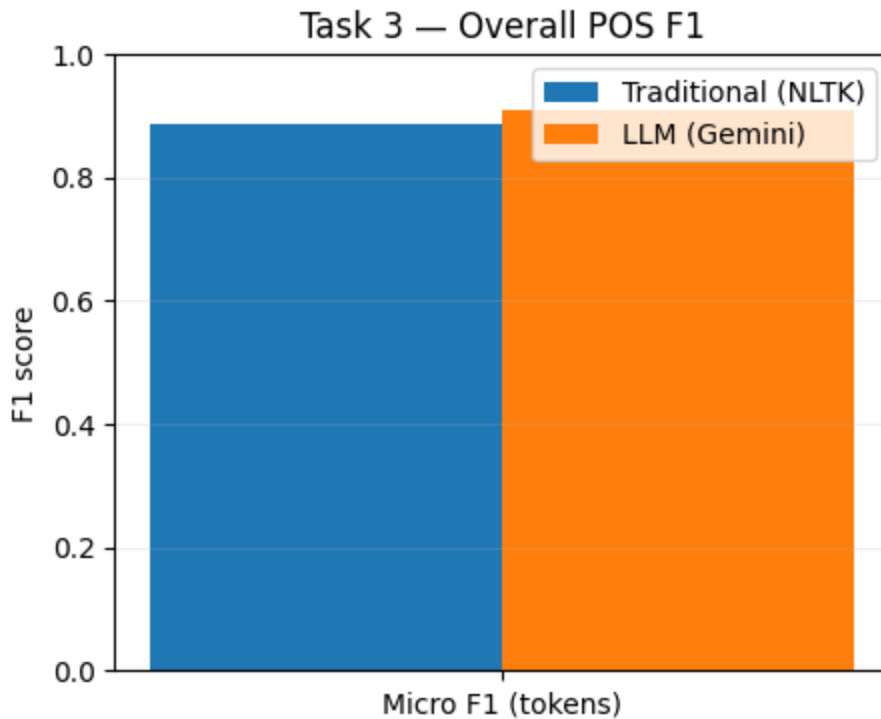
Use **NLTK** for tokenization and POS tagging: Tokenize using `nltk.word_tokenize` and tag using `nltk.pos_tag`. Convert predictions into "word/TAG" pairs and join into one string.

POS tags (top 12 by gold frequency)



Micro F1 (token-level) — Traditional (NLTK): 0.8863

Micro F1 (token-level) — LLM (Gemini): 0.91



Task 4: Topic Classification (AG News)

Dataset: AG News ([fancyzhx/ag_news](https://huggingface.co/datasets/fancyzhx/ag_news), Hugging Face)

Evaluation set construction (from test split):

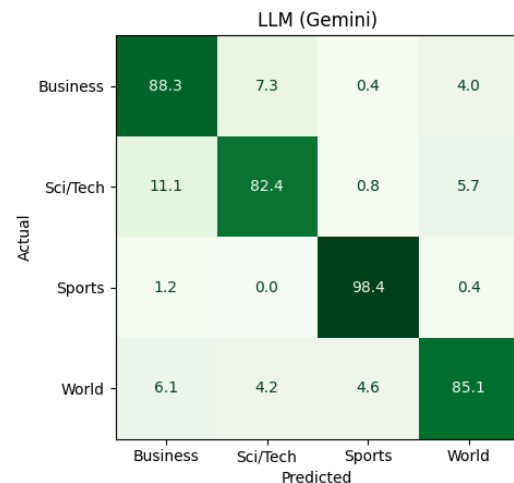
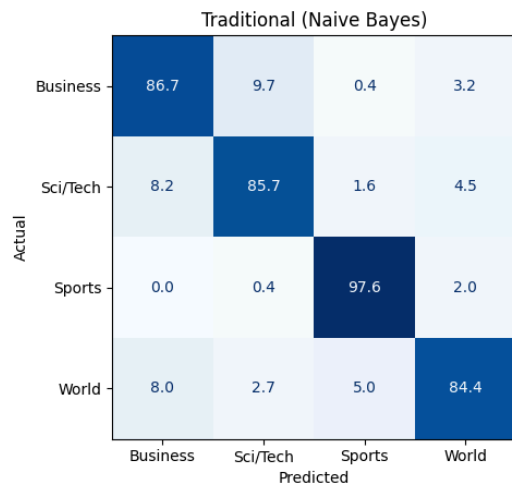
The dataset preparation for the evaluation involved mapping integer labels to specific topic names: 0 for World, 1 for Sports, 2 for Business, and 3 for Sci/Tech. A fixed seed was used to shuffle the test split deterministically, and a subset of 1,000 articles was selected for the

evaluation. The resulting data was stored with the article texts under `topic_data` and the corresponding topic names as `topic_ground_truth`.

Traditional baseline method (`traditional_nlp.py`):

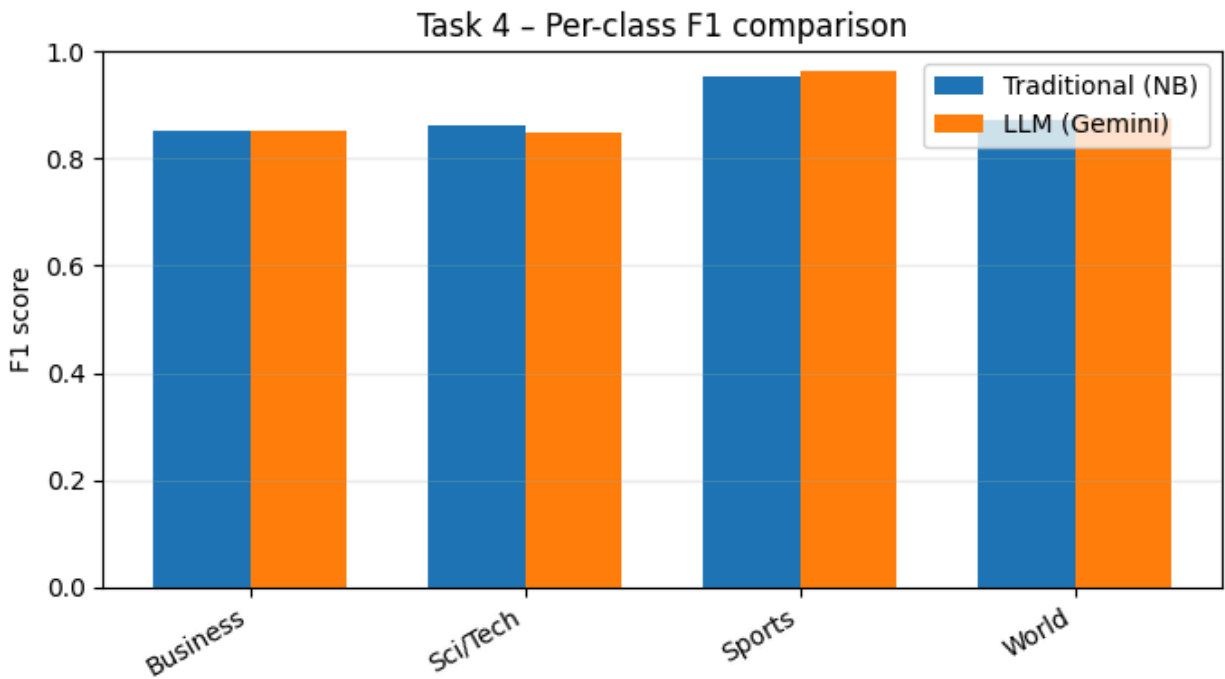
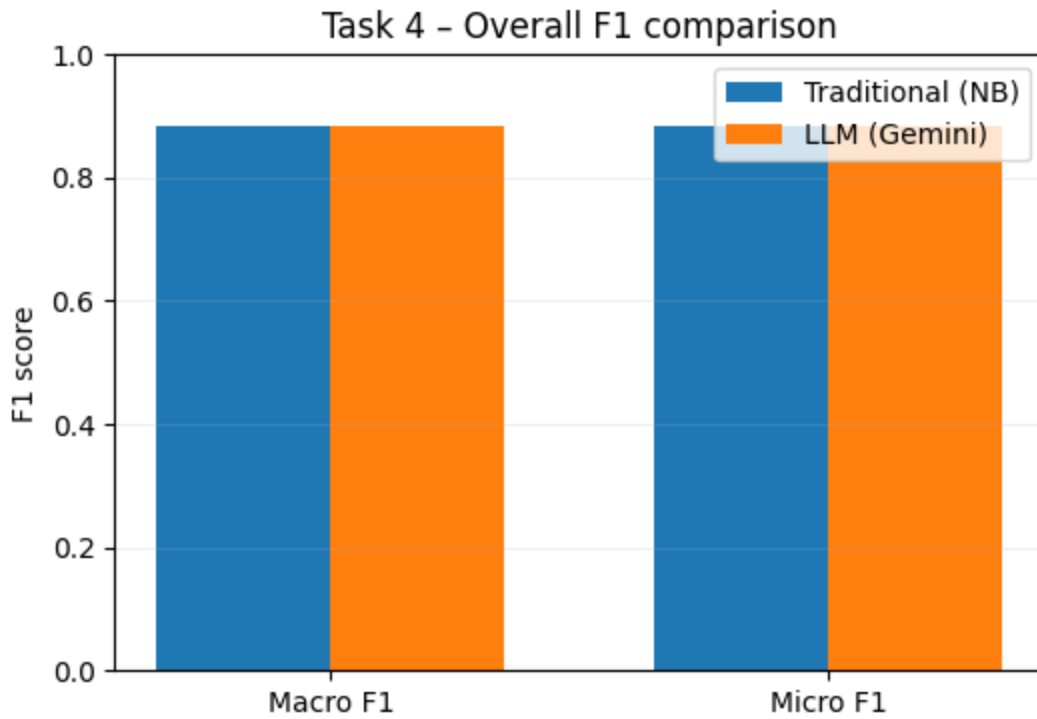
To train a classifier for the AG News topic classification task, the **TF-IDF + Multinomial Naive Bayes** approach is used. The process involves loading the AG News train split and applying the same label mapping, followed by vectorizing the texts using `TfidfVectorizer` with a maximum of 50,000 features and English stop words, and then training a `MultinomialNB` model on the resulting TF-IDF features. Evaluation entails transforming 1,000 evaluation texts with the fitted TF-IDF vectorizer, predicting the topic labels (as topic names), and comparing them to the `topic_ground_truth` to compute accuracy and record the correctness of each prediction.

Task 4: Topic Classification — Confusion Matrices (%)



Macro F1 (Traditional NB): 0.8848

Macro F1 (LLM Gemini): 0.8846



Task 5: Language Identification

Dataset: [papluca/language-identification](#) (Hugging Face)

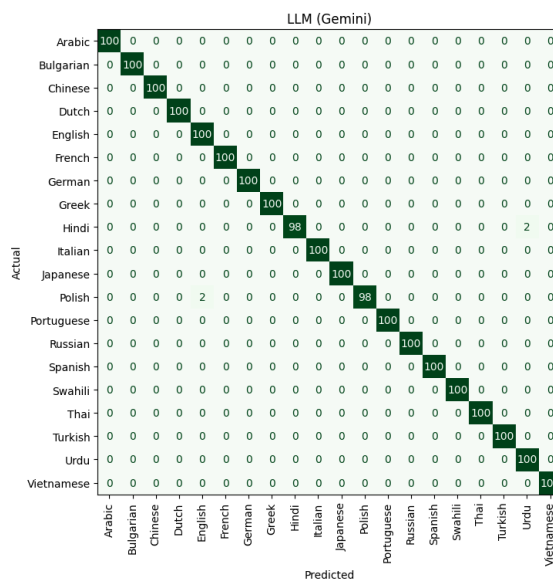
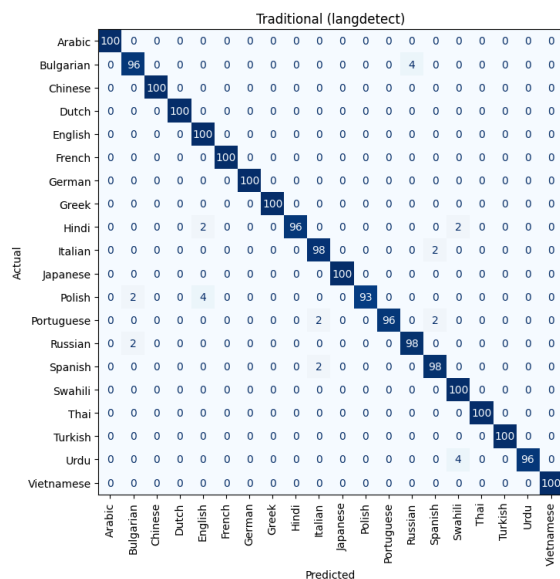
Data construction (from test split):

The **test split** was used, consisting of 10,000 samples across 20 languages. For each example, text was used as the input sentence or short passage. The ISO language code (e.g., "en," "fr") was mapped to a full language name using LANGID_CODE_TO_NAME. Deterministic shuffling was performed using a fixed seed, and 1,000 texts were selected. The texts were stored as [language_data](#), and the language names (20-class labels) were stored as [language_ground_truth](#).

Traditional baseline method:

Use **langdetect** to predict a language code for each text. Map predicted codes to the same 20 language names via LANG_MAP, including common fallbacks such as: "zh-cn" and "zh-tw" → "Chinese". Compare the predicted language name with the gold label and mark correctness.

Task 5: Language Identification — Confusion Matrices (%)



Traditional Macro F1: 0.9850

LLM (Gemini) Macro F1: 0.9979

